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Exp No: 4	Setup and configure a LAN using a Switch and Ethernet cables

AIM:

To set up and configure a LAN using a switch and Ethernet cables, assign IP addresses to the hosts, check the connectivity using the ping command and share files and folders across the machines of the LAN.

ALGORITHM / PROCEDURE:

1. Connect 4 host machines to the switch using straight-through UTP cables (RJ45). Check that the link LEDs of the switch glow.
2. Assign a static IP address to each host: **Windows**, Control Panel > Network and Sharing Centre > Change adapter settings > Ethernet > Properties > IPv4 (or run `ncpa.cpl`).
3. Use the addresses `192.168.10.11` to `192.168.10.14` with subnet mask `255.255.255.0`. Leave the default gateway empty because there is only one network.
4. Open the Command Prompt and verify the settings with `ipconfig`.
5. Check the connectivity between the machines with `ping`, for example from PC1 `ping 192.168.10.12`.
6. On PC1 create a folder `LabShare`, right click > Properties > Sharing > Advanced Sharing > tick **Share this folder** and give **Read/Write** permission to **Everyone**.
7. On PC2 open **File Explorer** and type `\\192.168.10.11` in the address bar. The files of PC1 are displayed.
8. The folder can also be mapped as a drive with `net use Z: \\192.168.10.11\LabShare` and the shared files can be copied between the machines.
9. Note down the output of each command.

Host	IP address	Subnet mask	Switch port
PC1	`192.168.10.11`	`255.255.255.0`	Fa0/1
PC2	`192.168.10.12`	`255.255.255.0`	Fa0/2
PC3	`192.168.10.13`	`255.255.255.0`	Fa0/3
PC4	`192.168.10.14`	`255.255.255.0`	Fa0/4

DIAGRAM:

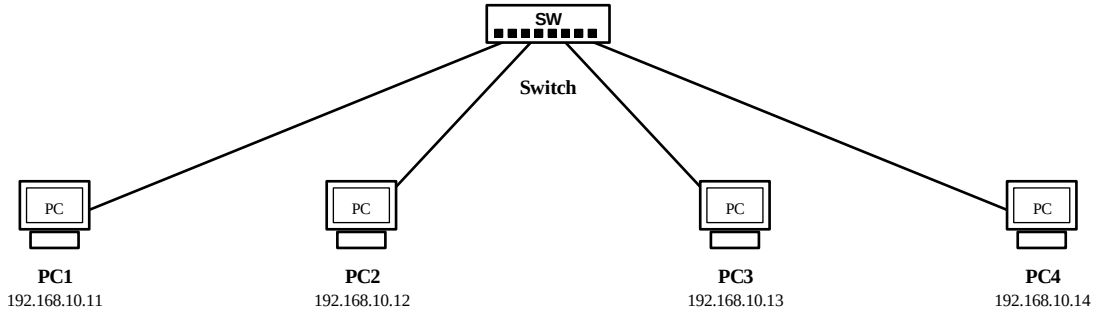


Fig 1: LAN with 4 hosts connected to a switch (star topology)

OUTPUT:

Command Prompt - PC1

```
C:\Users\UuUsSeErR>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6c1f:3a2e:91b4:d7c2%8
   IPv4 Address. . . . . . . . . . . : 192.168.10.11
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

Command Prompt - PC1

```
C:\Users\UuUsSeErR  >ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:
Reply from 192.168.10.12: bytes=32 time=1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\UuSsEeRr >ping 192.168.10.13

Pinging 192.168.10.13 with 32 bytes of data:
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128
Reply from 192.168.10.13: bytes=32 time=1ms TTL=128
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\UuSsEeRr >ping 192.168.10.14

Pinging 192.168.10.14 with 32 bytes of data:
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Command Prompt - PC2

```
C:\Users\uUsSeErR>net view \\192.168.10.11
Shared resources at \\192.168.10.11

Share name   Type   Used as   Comment
-------------------------------------------------------------------------------
LabShare     Disk
The command completed successfully.

C:\Users\UuUsSeErR>net use Z: \\192.168.10.11\LabShare
The command completed successfully.

C:\Users\uUsSeErR>dir Z:
 Volume in drive Z is LabShare
 Directory of Z:\

07-10-2026  08:42 PM    <DIR>          .
```

Command Prompt - PC2 (contd.)

```
07-10-2026  08:42 PM    <DIR>          ..
07-10-2026  08:40 PM            14,336 CN_Lab_Notes.docx
07-10-2026  08:41 PM           212,480 Exp4_Topology.png
07-10-2026  08:42 PM             1,024 hosts_list.txt
               3 File(s)        227,840 bytes
               2 Dir(s)  48,213,327,872 bytes free

C:\Users\UuUsSeErR   >copy Z:\hosts_list.txt C:\Users\UuUsSeErR   \Desktop
        1 file(s) copied.
```

RESULT:

Thus a LAN was set up with 4 hosts connected to a switch using Ethernet cables, the IP addresses were assigned, the connectivity was verified using the ping command and files were shared and accessed across the machines of the LAN.